CANONICAL EQUIVALENCE OF THE QUOTIENT STACK AND THE TORSOR QUOTIENT STACK

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1527-1536 of sources/stacks-examples-stacks/pinned-original-source.tex; proof span is lines 1538-1673, the complete author proof (134 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 0 of queue_examples-stacks.json (pb=4979, stmt 1527-1536, proof 1538-1673, halfA).

Editorial note. stacks-examples-stacks wave-1 (exs1): fresh adjudication accept (2026-10-14); pool per Chief 09:4x authorization (idle-line throughput over kun's backup note, SMC precedent); examples-stacks.tex 27 labels/20 with proof judged 9 accept / 11 reject; claim_key v2 stacks:examples-stacks.tex:<label>.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original proposition 0049: sources/stacks-examples-stacks/pinned-original-source.tex, source lines 1673–1673. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 04WL: sources/stacks-examples-stacks/pinned-original-source.tex, source lines 1418–1426. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 044Q: sources/stacks-examples-stacks/pinned-original-source.tex, source lines 1673–1673. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 044V: sources/stacks-examples-stacks/pinned-original-source.tex, source lines 1673–1673. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 04MB: sources/stacks-examples-stacks/pinned-original-source.tex, source lines 1673–1673. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 046N: sources/stacks-examples-stacks/pinned-original-source.tex, source lines 1673–1673. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0436: sources/stacks-examples-stacks/pinned-original-source.tex, source lines 1673–1673. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Proposition 0.1. *In Situation 04WL there exists a canonical equivalence*

$$[X/G] \longrightarrow [[X/G]]$$

of stacks in groupoids over $(Sch/S)_{fppf}$.

Proof. We write this out in detail, to make sure that all the definitions work out in exactly the correct manner. Recall that $[X/G]$ is the quotient stack associated to the groupoid in algebraic spaces $(X, G \times_B X, s, t, c)$, see Groupoids in Spaces, Definition 044Q. This means that $[X/G]$ is the stackification of the category fibred in groupoids $[X/_pG]$ associated to the functor

$$(Sch/S)_{fppf} \longrightarrow \mathit{Groupoids}, \quad U \longmapsto (X(U), G(U) \times_{B(U)} X(U), s, t, c)$$

where $s(g, x) = x$, $t(g, x) = a(g, x)$, and $c((g, x), (g', x')) = (m(g, g'), x')$. By the construction of Categories, Example 0049 an object of $[X/_pG]$ is a pair (U, x) with $x \in X(U)$ and a morphism $(f, g) : (U, x) \to (U', x')$ of $[X/_pG]$ is given by a morphism of schemes $f : U \to U'$ and an element $g \in G(U)$ such that $a(g, x) = x' \circ f$. Hence we can define a 1-morphism of stacks in groupoids

$$F_p : [X/_pG] \longrightarrow [[X/G]]$$

by the following rules: On objects we set

$$F_p(U, x) = (U, \pi \circ x, G \times_{B,\pi\circ x} U, a \circ (\mathrm{id}_G \times x))$$

This makes sense because the diagram

$$\begin{array}{ccccc} G \times_{B,\pi\circ x} U & \xrightarrow[\mathrm{id}_G \times x]{} & G \times_{B,\pi} X & \xrightarrow[a]{} & X \\ \downarrow & & & & \downarrow \pi \\ U & & \xrightarrow{\qquad\qquad \pi\circ x \qquad\qquad} & & B \end{array}$$

commutes, and the two horizontal arrows are G-equivariant if we think of the fibre products as trivial G-torsors over U, resp. X. On morphisms $(f, g) : (U, x) \to (U', x')$ we set $F_p(f, g) = (f, R_{g^{-1}})$ where $R_{g^{-1}}$ denotes right translation by the inverse of g. More precisely, the morphism $F_p(f, g) : F_p(U, x) \to F_p(U', x')$ is given by the cartesian diagram

$$\begin{array}{ccc} G \times_{B,\pi\circ x} U & \xrightarrow[R_{g^{-1}}]{} & G \times_{B,\pi\circ x'} U' \\ \downarrow & & \downarrow \\ U & \xrightarrow{\quad f \quad} & U' \end{array}$$

where $R_{g^{-1}}$ on T-valued points is given by

$$R_{g^{-1}}(g', u) = (m(g', i(g(u))), f(u))$$

To see that this works we have to verify that

$$a \circ (\mathrm{id}_G \times x) = a \circ (\mathrm{id}_G \times x') \circ R_{g^{-1}}$$

which is true because the right hand side applied to the T-valued point (g', u) gives the desired equality

$$\begin{aligned} a((\mathrm{id}_G \times x')(m(g', i(g(u))), f(u))) &= a(m(g', i(g(u))), x'(f(u))) \\ &= a(g', a(i(g(u)), x'(f(u)))) \\ &= a(g', x(u)) \end{aligned}$$

because $a(g, x) = x' \circ f$ and hence $a(i(g), x' \circ f) = x$.

By the universal property of stackification from Stacks, Lemma 0436 we obtain a canonical extension $F : [X/G] \to [[X/G]]$ of the 1-morphism F_p above. We first prove that F is fully faithful. To do this, since both source and target are stacks in groupoids, it suffices to prove that the *Isom*-sheaves are identified under F. Pick a scheme U and objects ξ, ξ' of $[X/G]$ over U. We want to show that

$$F : \mathit{Isom}_{[X/G]}(\xi, \xi') \longrightarrow \mathit{Isom}_{[[X/G]]}(F(\xi), F(\xi'))$$

is an isomorphism of sheaves. To do this it suffices to work locally on U, and hence we may assume that ξ, ξ' come from objects (U, x), (U, x') of $[X/_pG]$ over U; this follows directly from the construction of the stackification, and it is also worked out in detail in Groupoids in Spaces, Section 04MB. Either by directly using the description of morphisms in $[X/_pG]$ above, or using Groupoids in Spaces, Lemma 044V we see that in this case

$$\mathit{Isom}_{[X/G]}(\xi, \xi') = U \times_{(x,x'),X \times_S X,(s,t)} (G \times_B X)$$

A T-valued point of this fibre product corresponds to a pair (u, g) with $u \in U(T)$, and $g \in G(T)$ such that $a(g, x \circ u) = x' \circ u$. (Note that this implies $\pi \circ x \circ u = \pi \circ x' \circ u$.) On the other hand, a T-valued point of $\mathit{Isom}_{[[X/G]]}(F(\xi), F(\xi'))$ by definition corresponds to a morphism $u : T \to U$ such that $\pi \circ x \circ u = \pi \circ x' \circ u : T \to B$ and an isomorphism

$$R : G \times_{B, \pi \circ x \circ u} T \longrightarrow G \times_{B, \pi \circ x' \circ u} T$$

of trivial G_T-torsors compatible with the given maps to X. Since the torsors are trivial we see that $R = R_{g^{-1}}$ (right multiplication) by some $g \in G(T)$. Compatibility with the maps $a \circ (1_G, x \circ u), a \circ (1_G, x' \circ u) : G \times_B T \to X$ is equivalent to the condition that $a(g, x \circ u) = x' \circ u$. Hence we obtain the desired equality of *Isom*-sheaves.

Now that we know that F is fully faithful we see that Stacks, Lemma 046N applies. Thus to show that F is an equivalence it suffices to show that objects of $[[X/G]]$ are fppf locally in the essential image of F. This is clear as fppf torsors are fppf locally trivial, and hence we win. □